



FORMAL SDLC REVIEW AND SPRINT RECORD

ZENTRA SDLC VERDICT AND EVIDENCE-BASED SPRINT RECORD

Verified implementation, open gates, and conditional backlog

Final SDLC Verdict

Hybrid Agile SDLC: Scrum-led delivery, DevSecOps engineering practices, and formal release gates. The present implementation is substantial, but production release is not authorized by the available evidence.

SPRINT 9

CURRENT SCHEDULE STATE

Evidence verified through 16 July

NOT READY

RELEASE STATE

Production blockers remain open

NOT ASSERTED

READINESS PERCENTAGE

No approved acceptance baseline

FORMAL DELIVERY WINDOW

19 April 2026 - 17 August 2026

April 19 is the declared start; the first repository commit is dated April 29.

Executive Evidence Statement

Current formal status

The application builds and its focused automated checks pass. The production database is reachable and current. However, production environment validation fails, no dedicated admin mobile/PWA artifact exists, and formal device, security, UAT, rollback, and handover evidence was not found.

Verified on 16 July 2026

- TypeScript checking and the optimized Next.js build completed successfully.
- Nineteen focused automated tests passed: booking orchestration, task templates, Command Center foundations, and export utilities.
- Lint completed with zero errors and 37 warnings.
- The production PostgreSQL database connected successfully; 30 migrations are applied and pgvector is available.
- The repository contains 211 API route files, 40 page files, and 56 Prisma models.

Verified blockers or missing evidence

- Production environment validation fails because NEXTAUTH_URL is not HTTPS and three required integration/security variables are missing.
- No web-app manifest, service worker, native mobile project, or mobile runtime dependency is present outside dependencies/build output.
- Responsive implementation patterns exist, but a recorded device acceptance matrix was not found.
- Signed security/privacy review, business UAT, backup/restore proof, rollback rehearsal, and operational handover were not found.

Percentage policy

A production-readiness percentage is not reported because the repository does not contain an approved, weighted acceptance baseline. Inventing a percentage would convert incomplete evidence into a false measure.

Evidence Rules and System Profile

Claim classifications used in this report

CLASSIFICATION	MEANING
Verified	Directly supported by a command result, file, or dated commit.
Observed, not accepted	Implementation exists; formal acceptance evidence is absent.
Failed	A required validation command returned a failure.
Not evidenced	No supporting artifact was found; no claim of completion is made.
Conditional backlog	Future required work; completion is not predicted or implied.

Current repository profile

AREA	REPOSITORY EVIDENCE
Application	Next.js 16.2.10, React 19.2.4, TypeScript 5, Tailwind CSS 4
Identity	NextAuth 4.24.14 with role-based administrative flows
Data	Prisma 7.8.0, PostgreSQL/Supabase, pgvector
Scope indicators	211 API routes; 40 pages; 56 data models; 30 migrations
Operations	n8n, email workflows, storage, Vercel/Render configuration

Limit of Git evidence

A commit proves that a change entered version control. It does not prove production deployment, usability, security, business acceptance, or completion. Empty commit windows mean only that this repository has no dated Git evidence for those periods.

Dated Repository Timeline

Verified dates and commit subjects

DATE	REPOSITORY RECORD	SUPPORTED INTERPRETATION
19 Apr	Declared project start	No repository commit in this interval
29 Apr	3a4df8ce - first commit	Initial Next.js repository baseline
9 Jun	385ae885 - frontend features	Public pages and booking-related UI
12 Jun	60b6fcc2 - admin panel features	Administrative interface expansion
19 Jun	70cd1f48 - database and admin role	Data and administrative identity
21 Jun	0940095a - login and CRUD/RBAC	Authentication and access-control reconstruction
27 Jun	449d0155 - architecture/orchestration	Architecture cleanup and orchestration
4 Jul	ee1b460d - architecture cleanup	Repository stabilization activity
9 Jul	f3b69c16 and related commits	Admin export implementation and cleanup
14 Jul	7876d7b8 - Command Center/deployment	Command Center and production configuration
15 Jul	Five deployment/workflow commits	Render configuration, Gmail OAuth, worker tuning

Timeline limitation

No commits were found in the May 9-June 7 sprint windows. This report leaves those backlog cells blank and records the evidence gap rather than inventing completed work.

Live Verification Results

Commands and repository checks

CHECK	EVIDENCE	STATUS
TypeScript	npm run typecheck	Passed
Focused tests	19 tests across four suites	Passed
Production build	Next.js optimized build; 38 generated pages	Passed
Lint	0 errors; 37 warnings	Passed with warnings
Production database	Connected; 30 migrations current; pgvector enabled	Passed
Production environment	HTTPS URL invalid; three required variables missing	Failed
Responsive web code	Breakpoint/overflow patterns observed in admin UI	Observed
Admin mobile/PWA	No manifest, service worker, native project, or runtime	Not present
Device acceptance	No recorded supported-device test matrix found	Not evidenced
UAT/release acceptance	No signed UAT, rollback, or handover record found	Not evidenced

Formal interpretation

The codebase has verified implementation strength. The failed production configuration and missing mobile, device, security, UAT, and operational acceptance evidence prevent a production-ready declaration.

Open Production Release Gates

Evidence still required before release

GATE	OBJECTIVE CLOSURE EVIDENCE	CURRENT STATE
Environment	Correct HTTPS URL and provide required webhook/scanner variables	Failed
Admin mobile	Committed client artifact with authenticated core workflows	Not present
Device support	Recorded phone, tablet, laptop, and desktop results	Not evidenced
Security/privacy	Approved review of RBAC, sessions, uploads, secrets, and data use	Not evidenced
Critical-path QA	Integrated booking, payment, contract, email, and failure-path tests	Open
Business UAT	Named users, scenarios, defects, retest, and signed acceptance	Not evidenced
Operations	Monitoring, backup/restore proof, rollback rehearsal, support ownership	Not evidenced
Handover	Runbooks, training record, release notes, and owner approval	Not evidenced

No percentage substitution

This gate register shows exactly what is passed, failed, present, or missing. It is more defensible than assigning an arbitrary percentage to incomparable evidence. The target remains 100% gate closure, not a forecasted score.

SDLC Verdict and Operating Model

Lifecycle model fit

MODEL	FIT	EVIDENCE-BASED ASSESSMENT
Waterfall	Low	Dated evidence shows changing scope and staged integration.
Pure Scrum	Partial	Iteration fits, but release and operational gates remain necessary.
Pure DevOps	Partial	Automation helps, but it does not replace backlog and UAT governance.
Hybrid Agile	Best fit	Combines short delivery cycles with explicit release evidence.

Final verdict: Hybrid Agile SDLC

Use Scrum for backlog control, sprint review, and stakeholder acceptance. Use DevSecOps for repeatable verification and deployment. Use formal gates for mobile acceptance, security/privacy, production configuration, UAT, rollback, and handover.

Required operating rules

- Every completed claim must link to a commit, test output, deployment record, or signed acceptance artifact.
- Future sprint rows are backlog commitments only; they do not imply that completion will occur by the date shown.
- Sprint reviews update gate states using evidence and record unresolved defects without re-labeling them as complete.
- The Product Owner may authorize release only when every mandatory gate has acceptable evidence.

Evidence-Based Sprint Record - Sprints 1-6

Past rows use dated repository evidence. Blank backlog cells indicate that no valid dated Git evidence was found. Future rows are explicitly conditional and do not claim predicted completion.

SPRINT	BACKLOG / VERIFIED FOCUS	EVIDENCE STATE	START	END
1 - Project kickoff	Formal start declaration	Declared; no Git commit	Apr 19	Apr 28
2 - Initial repository baseline	Next.js application scaffold	Verified: 3a4df8ce	Apr 29	May 8
3 - Evidence gap		No dated Git evidence	May 9	May 18
4 - Evidence gap		No dated Git evidence	May 19	May 28
5 - Evidence gap		No dated Git evidence	May 29	Jun 7
6 - Frontend and admin UI	Public pages; booking; admin panel	Verified: Jun 9 and 12	Jun 8	Jun 17

Evidence note

The formal project start is retained as 19 April 2026. The first dated repository commit is 29 April. No unsupported deliverables are assigned to the intervening empty Git windows.

Evidence-Based Sprint Record - Sprints 7-12

Past rows use dated repository evidence. Blank backlog cells indicate that no valid dated Git evidence was found. Future rows are explicitly conditional and do not claim predicted completion.

SPRINT	BACKLOG / VERIFIED FOCUS	EVIDENCE STATE	START	END
7 - Data, auth, RBAC, orchestration	Database; admin roles; CRUD/RBAC	Verified: Jun 19-27	Jun 18	Jun 27
8 - Architecture cleanup	Repository cleanup and stabilization	Verified: Jul 4	Jun 28	Jul 7
9 - Exports and Command Center	Exports; deployment; email workflow	Active; verified to Jul 16	Jul 8	Jul 17
10 - Admin mobile delivery	Choose client approach; core workflows	Required; not started	Jul 18	Jul 27
11 - Cross-platform validation	Device tests; security; UAT	Required; not started	Jul 28	Aug 6
12 - Release acceptance	Environment; rollback; handover	Required; not started	Aug 7	Aug 17

Current position - Sprint 9

Repository activity through 16 July verifies export, Command Center, deployment, and email-workflow changes. No mobile client artifact is present, so mobile work is not recorded as completed or active evidence in this sprint.

Admin Mobile and Adaptive Web Reality

Verified position

The web application contains responsive layout patterns and a broad secured API surface. A dedicated admin mobile application or installable PWA is not present. The report therefore treats mobile delivery as an open requirement, not an existing capability.

What is present and what is absent

ITEM	REPOSITORY EVIDENCE	STATE
Responsive web styles	Present in public and admin components	Observed, not device accepted
Shared API routes	211 route files cover public, client, admin, and orchestration	Present
Mobile web manifest	No manifest.webmanifest found	Absent
Service worker	No application service worker found	Absent
Native mobile project	No Android, iOS, Expo, React Native, or Capacitor project	Absent
Device test record	No supported-device acceptance matrix found	Not evidenced

Required mobile delivery order

- Approve the mobile delivery approach and supported device/browser list before implementation.
- Create the client shell and verify authentication, role enforcement, session expiry, and sign-out.
- Implement only prioritized operational workflows: dashboard, bookings, inquiries, calendar, tasks, and alerts.
- Record phone and tablet results, security checks, failures, fixes, and Product Owner acceptance.
- Keep complex configuration, bulk operations, and high-density reports web-first until separately accepted.

Objective Completion Conditions

Evidence required for the 100% production-ready claim

CONDITION	MINIMUM OBJECTIVE EVIDENCE
Production configuration	Environment validator passes with HTTPS and all required variables
Admin mobile artifact	Committed client builds and authenticates against secured APIs
Cross-device acceptance	Recorded results for supported phones, tablets, laptops, and desktops
Integrated QA	Critical booking, payment, contract, email, retry, and failure paths pass
Security/privacy	Named reviewer approves RBAC, sessions, uploads, secrets, and data handling
Business UAT	Named users complete scenarios; defects are closed or formally accepted
Operational readiness	Monitoring, backup/restore, rollback, ownership, and escalation are proven
Handover	Runbooks, release notes, training, and Product Owner approval are signed

Date rule

17 August 2026 is the stated target date. It is not evidence of completion. If any mandatory condition remains failed, absent, or not evidenced on that date, the truthful status remains not production-ready and the unresolved item must be carried forward.

Final Realistic Verdict

Formal conclusion

Hybrid Agile remains the appropriate SDLC. The system has a substantial verified web implementation, passing focused checks, and a current production database. It is not production-ready because the environment validation fails and required mobile, device, security, UAT, rollback, and handover evidence is absent.

What can be stated now

- The project is formally dated from 19 April 2026; version-control evidence begins on 29 April.
- Sprint 9 is active by schedule, with repository evidence through 16 July.
- Build, type checking, 19 focused tests, lint execution, and production database checks have current evidence.
- The admin mobile deliverable and cross-platform acceptance remain open requirements.
- No defensible production-readiness percentage can be asserted from the available acceptance evidence.

What remains before release

- Resolve the failed production environment validation.
- Commit and verify the selected admin mobile client and its prioritized workflows.
- Complete recorded device, security, integrated QA, and business UAT evidence.
- Prove monitoring, backup/restore, rollback, support ownership, and handover.

FINAL VERDICT

Proceed with Hybrid Agile governance and the evidence-classified 12-sprint record. Keep future rows conditional, preserve blank backlog cells where evidence is absent, and declare 100% only after every mandatory release condition has objective proof and formal acceptance.